

Exploring Learning Trajectories in Collaborative Educational Games

Abstract

This paper studies how learners develop computational literacy in playing a 2-player collaborative constructionist game through investigating the play logs. The paper builds on the work in play-tracing and game learning analytics. We present data from 128 sessions with 3 types of learning outcome: play-based (n=31), goal-based (n=20), and general play (n=77). The results showed that players develop their knowledge of the game through **Explore, Tinker and Refine** (ExTiRe), a learning framework that shows how novice programmers learn to program (Berland et al., 2014). The study contributes to the field by adding empirical evidence on how learners learn in playing constructionist game. Additionally, it presents a scalable and reproducible method in investigating learning games.

1. Objectives or Purposes

Work in learning analytics and educational data mining (Baker & Siemens, 2014) has identified a host of methods for analysis of log data from game systems, thus enabling insights applicable to teaching computational literacy (Berland et al., 2014). Owen & Baker (2020) used relevant learning analytics methods including visualization, structure discovery, relationship mining, and prediction, showing how these processes can support deep insights into learning patterns in games.

This study investigates **how play-tracing can effectively uncover player learning pathways** to provide insights on fostering computational literacy in a 2-player collaborative learning game currently exhibited in two science museums.

Imagining better futures calls on us to design tools and methods that actively broaden participation instead of narrowing it. This study advances that agenda by illuminating how learners from diverse backgrounds engage with core computational ideas through playful interaction in a museum game. We also introduce an analytic approach that turns messy, unstructured gameplay logs into actionable insights—making it more practical to monitor, support, and compare learning trajectories at scale.

2. Theoretical Framework

Constructionism posits that knowledge needs to be constructed to be understood (Papert, 1993), and games invite learners to construct the knowledge by building ideas, perceptions, and artifacts that are meaningful to learners. In constructionist games, learners solve game challenges through form, develop, build, test, and refine their knowledge about the games (Holbert, 2019). Therefore, constructionism has been shown to be particularly well suited for introducing computer science to novices (Berland & Wilensky, 2015; Byun et al., 2025).

One challenge to understand and analyze the learning from games lies in the complexity of the unstructured game telemetry. Game Learning Analytics (GLA) involves a systematic measurement, collection, analysis, and visualization of the fine-grained interaction data produced while people are playing games (Calvo-Morata et al., 2024). A typical goal is to generate evidence about what, how and to what extent people are learning, thus providing insights for researchers, game designers, instructors so that the game can be adapted and improved, and learning engagement and trajectory optimized (Berland et al., 2014). Previous studies have explored how GLA can be used to reveal the process of learning. For example, Berland et al. (2014) found that novice learners of computer science demonstrated the process of Explore, Tinker, and Refine (ExTiRe) in learning programming in a constructionist game designed for

learning computer science. In our study, we examine if the ExTiRe learning framework persists in a 2-player collaborative game. Andersen et al. (2010) has explored dimensionality-reduction techniques in identifying play strategies and finding areas of confusion to inform iterative game design by applying multidimensional scaling of “player state” to visualize players’ game trajectories; they call this “play-tracing,” and it has been used by to improve the design of learning games (e.g., Berland et al., 2014; Horn et al., 2016) . That said, current literature on play-tracing has relatively little exploration of *player learning trajectories* or *group behavior*, both core ideas in game learning analytics. **This paper builds on work in play-tracing to better understand how to see and use player learning trajectories in game learning analytics; this may enable us to both better understand how students learn differently in games and better design learning games in the future.**

3. Methods

3.1 Rainbow Agents

Rainbow Agents is a museum exhibit game designed to support the development of computational literacies through agent-based modeling (ABM), which involves using computational techniques to manage individual agents in a larger system.

In the game, there are three programmable creatures, each representing a unique computational modeling idea (see Figure 1). These creatures can take actions such as planting different types of plants, watering, on assigned locations in the garden following the certain computational logic that they represent. Learners choose the programmable characters, select a territory (a 3 x 3 tile in the garden), decide the type of cards they want, and arrange the sequence to program the characters.

If you successfully create working code, you fill a relevant “Orb.” There are three types of orbs in the game (mushroom, woody, and leafy). When you fill all three orbs, the garden rains. Rain makes the shared garden healthier, enabling learners to find treasure and special rainbow plants (see Figure 2). Players can play indefinitely, but all “game-directed goals” are met when players are opening treasure chests and planting rainbow plants. Filling orbs and opening treasure chests each need a different set of game tactics.

3.2 Data Source

We analyzed gameplay logs from a random sample of 128 sessions of open museum play with *Rainbow Agents*. Each session was treated as a single episode of gameplay, as each visitor in the sample played the game only once. The game automatically logged time-stamped sequences of all in-game actions.

To compare group learning behavioral patterns, we categorize groups with learning outcomes into play-based achievement (ACH_PLAY, n=31 sessions), goal-based achievement (ACH_GOAL, n=20), and general play (ACH_NONE, n=77), yielding 78,575 lines of game interaction data. ACH_PLAY could be thought of as “having many small wins” – in, Super Mario Brothers, this would be equivalent to “getting many coins but not finishing a level”; ACH_GOAL would be players who finished levels or completed the game; ACH_NONE would be those players who “played around” but did not collect many coins or finish a level.

3.3 Play-tracing

We operationalized ExTiRe behaviors through four core game mechanic dimensions (See Table 1.) These features were computed at the state (log line) level, representing discrete player actions over time.

3.4 Learning Outcome Constructs

Each player's learning outcome was inferred from their in-game progress toward key goal states, operationalized into three categories:

- **ACH_PLAY** – task-relevant game orbs were successfully activated,
- **ACH_GOAL** – the treasure chest was opened (an advanced game feature),
- **ACH_NONE** – neither of the above goals was met.

These outcomes were not explicitly required by the game, yet they represent escalating levels of mastery and are used as proxy indicators for achievement in the game (and, as such, success in learning to write working code in game).

3.5 Temporal Analysis

We examined whether players exhibited ExTiRe-consistent learning trajectories by segmenting sessions into *Early*, *Middle*, and *Late* phases using the total number of session data from each session. Based on ExTiRe theory, we hypothesized that effective learners would demonstrate high early exploration (Navigation) in the early phase, and drop as they play, showing a more focused and efficient strategy use. In ExTiRe, tinkering is essential for learning to program, as players try different play tactics and learn through making productive trials before they refine the strategies. Therefore, we hypothesize that learners may show a medium to high level of experimental engagement (Tactics) in early and middle phase and become more capable of integrating and synthesizing strategies in the late phase. As planting is one primary intermediate milestone toward the larger achievement of the game, we hypothesized that data would show an increase of successful planting or a high level of planting throughout the gameplay (Planting). Lastly, similar to navigation, as learners play, we hypothesize players will

adapt more efficiently, therefore, it is likely to see a drop of execution cost but an increase in planting and tactics (Entropy).

[Table 2]

We examine both within-group behavioral evolution patterns and between-group phase-specific differences using non-parametric statistical methods to account for non-normal behavioral score distributions. Because of space constraints, we skip the discussion of between-group results in this paper.

4. Results and/or substantiated conclusions

The study investigates how play-tracing uncover player learning pathways and provide insights on learning in a collaborative learning game. Specifically, we investigate: **Are there any temporal patterns across distinct learning outcome groups? Did players develop computational literacy through an ExTiRe trajectory?**

To examine potential shifts in player behavior over the course of gameplay, we analyzed core activity patterns within distinct temporal phases of the game using play-tracing. We employed both within-group and between-group analyses to capture different dimensions of behavioral change. While within-group analysis reveals how behavior evolves internally within each outcome group, it may obscure broader trends. To address this, we complement it with between-group comparisons, which highlight how behavioral trajectories diverge across groups over the course of gameplay.

[Table 3]

The results from the two groups achieved major game accomplishments (ACH_PLAY & ACH_GOAL) showed that learners developed computational literacy through **Explore, Tinker and Refine** learning pathways. (See Figure 3, Figure 4) Particularly, in the early phase of

gameplay, all three groups exhibited a high level of navigation activities (Navigation), showing active **exploration** of the game. As learners play, we observed a significant amount of drop in navigation, and coupled with an increase in planting outcome. Although we saw all groups scaling down spatial exploration as they play, the ACH_GOAL group showed a relatively high amount of navigation in the middle phase to their counterparts. To open treasure chests, players need to figure out the right places to plant, therefore, the residue amount of high movement suggested an active **tinkering** process experimenting with a smaller set of hypotheses in figuring out the strategy.

Our findings suggested that learners both experimented with their codes and made progressive **refinement** of their game tactics, thus providing empirical support for the ExTiRe learning pathway. Specifically, the ACH_GOAL group exhibited a Low–Low–High trajectory in Tactics and Planting, indicating a prolonged experimentation phase and eventually arriving at an effective strategy that enabled them to open the chests, and the ACH_PLAY group followed a U-shaped High–Low–Medium pattern in play tactics. (See Figure 3) This suggests a trajectory from initial exploration with a variety of things (early success), to forming specific hypotheses, testing codes with trial and errors (mid-phase decline), to final refinement (eventual improvement).

Additionally, both groups showed a progression from a low to high complexity of tactics in gameplay, demonstrating a development of understanding more complex computational logics and applying them to achieve their game objectives. (See Figure 3) This mastery of computational literacy is also reflected in the decrease of Entropy levels, where we see players being able to use fewer cards to plant more, suggesting a more efficient coding execution rate compared to the early phase where players use more cards to achieve the same level of plants. This transition from a usage of simple computational logic to a more efficient complex one

showed how players **refined** their play strategies with the development of expertise in different computational logics in the game.

5. Scientific Significance

This work contributes to learning analytics and learning sciences by introducing an analytical pipeline that enables researchers, game designers and instructors to analyze and interpret the complex, non-linear, and unstructured game interactions from learning games using data mining techniques. It provides researchers with a tool to combine trajectory embedding with player states. This allows researchers to conduct exploratory between-group analyses with a large dataset. This framework addresses a critical methodological gap in analyzing how to systematically analyze complex, unstructured collaborative game interactions while preserving both group learning pathways and temporal dynamics.

Also, the study provides evidence for fostering computational literacy in museums; visitors (mostly children) played the game only for a few minutes, but they were able to build working agent-based models that achieved local goals. In using our method, we found that when playing in a 2-player collaborative museum game, learners demonstrated the “Explore, Tinker, and Refine” learning trajectory, extending the findings from previous studies on how learners develop computational literacy in playing constructionist games.

Informal learning environments, such as local science museums, can lower the barriers faced by traditionally underserved communities to get access to and study complex computer science concepts (e.g. parallelism, concurrency). Though many studies show that visitors largely have “entertainment” focused experiences (Lepouras & Vassilakis, 2004), this study shows that their experiences can include relatively complex learning trajectories of academic content, similar to in-school games.

Appendix

Table 1. Operationalize ExTiRe in Rainbow Agents

Dimension	Description	Specific Features
Navigation	Spatial exploration	Game interactions related to navigation
Tactics	The mental model of how they can operate the characters through computational modeling	Game interactions show how players use different cards
Planting	Main instant outcome	Game interactions related to planting actions
Entropy	Instant feedback from the system that shows implementation effectiveness	Game interactions reflect the instant results of the actions

Table 2. Hypotheses of Feature Dimension Tread with Gameplay Progression

	Early	Middle	Late
Navigation	High	Medium	Low
Tactics	Medium/High	High	High
Planting	Medium/High	Low/Medium	High
Entropy	Medium/High	Medium	Low

Table 3. Overview of All Groups Trend across Gameplay

	Early			Middle			Late		
	ACH_PLAY	ACH_GOAL	ACH_NONE	ACH_PLA_Y	ACH_GOAL	ACH_NONE	ACH_PLA_Y	ACH_GOAL	ACH_NONE
Navigation	High	High	High	Medium	Medium	Low	Low	Low	Low

Tactics	High	Low	Medium	Low	Low	High	Medium	High	Low
Planting	Low	Low	Low	High	Low	High	High	High	High
Entropy	High	High	Medium	Medium	Medium	High	Low	Low	Low

Figure 1. The Three Creatures in Rainbow Agents

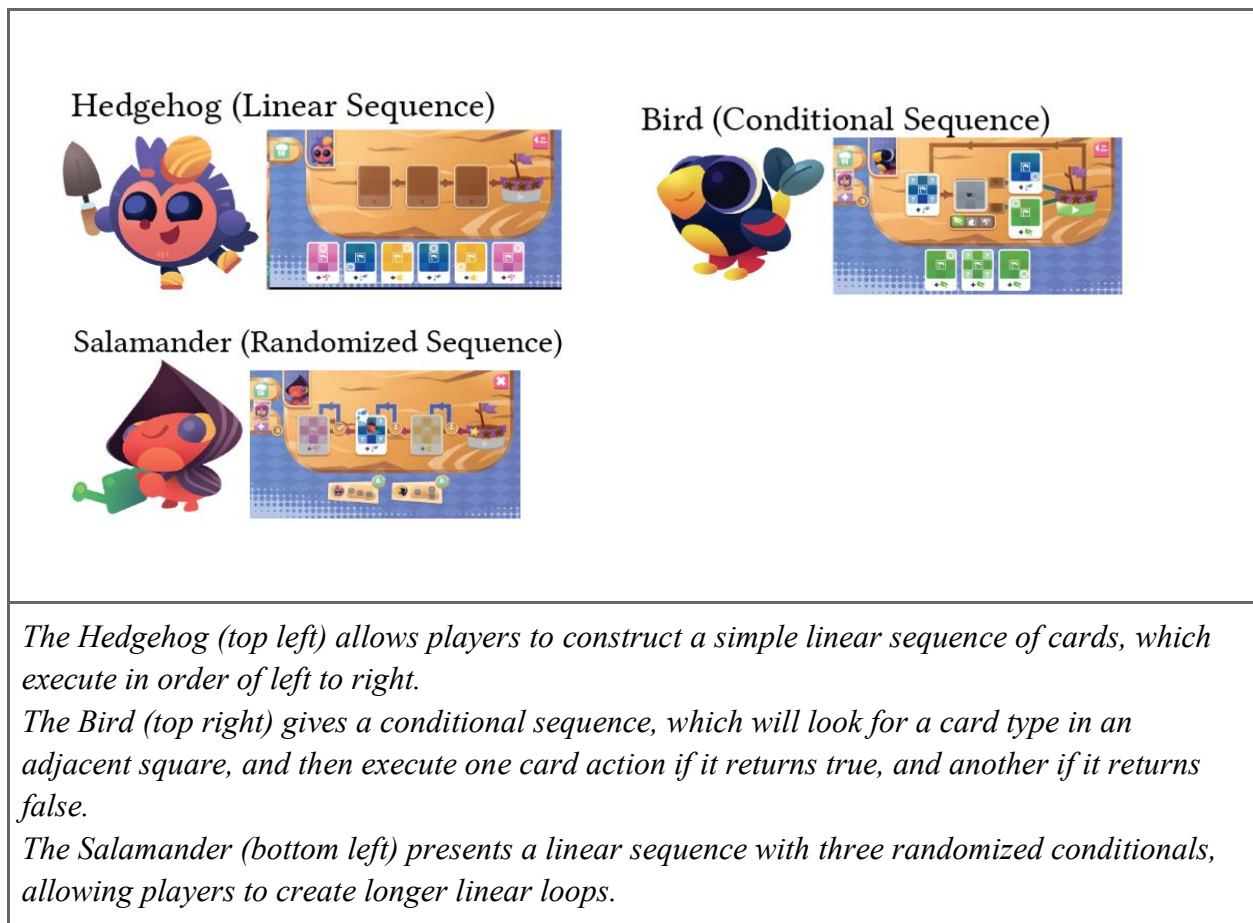


Figure 2. The Shared Garden in Rainbow Agents with Orbs and Treasure



Figure 3. Trend Plots of Behavioral Dimensions Across Groups

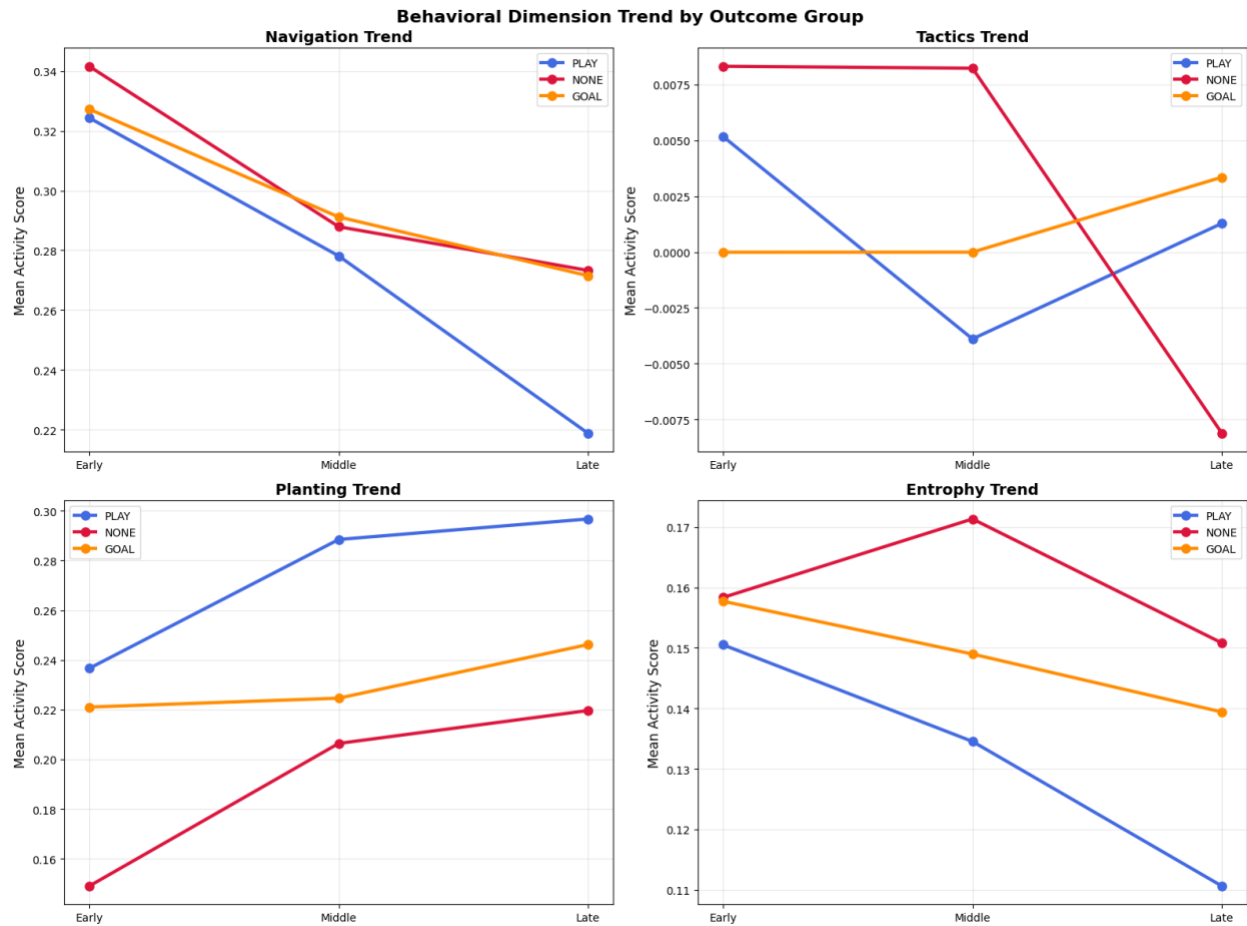
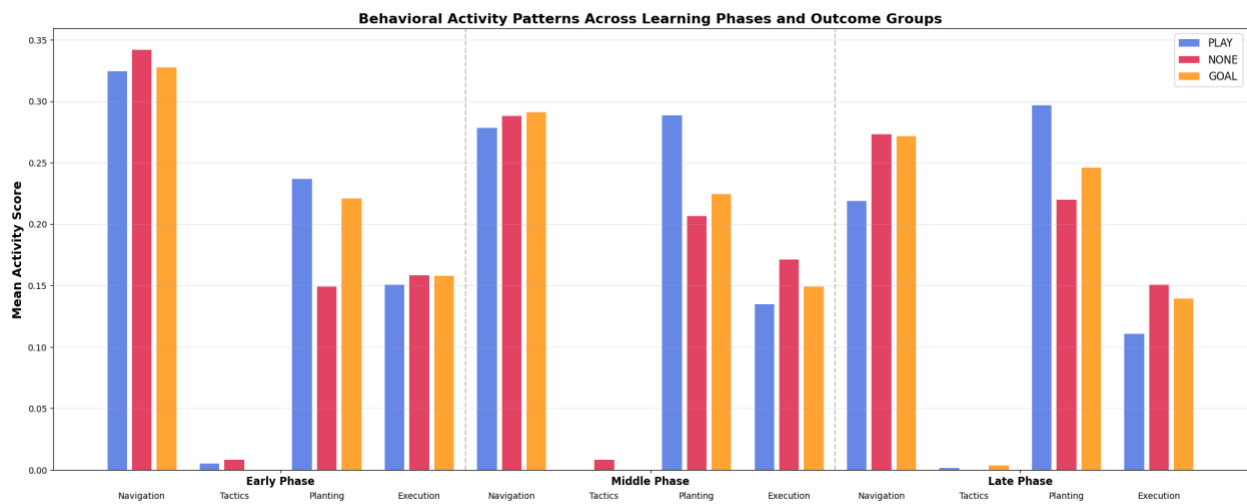


Figure 4. Behavioral Activity Patterns Across Learning Phases and Outcome Groups



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